Getting Started in Bug Bounty: 10 Non-obvious S3 Integration Vulnerabilities in Web Applications (English translation)

Начинаем в багбаунти: 10 неочевидных уязвимостей S3-интеграций в веб-приложениях - Sergey Bobrov (BlackFan), Positive Technologies / Habr.

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Content (translated into English)

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Positive Technologies

4.21 Employer rating

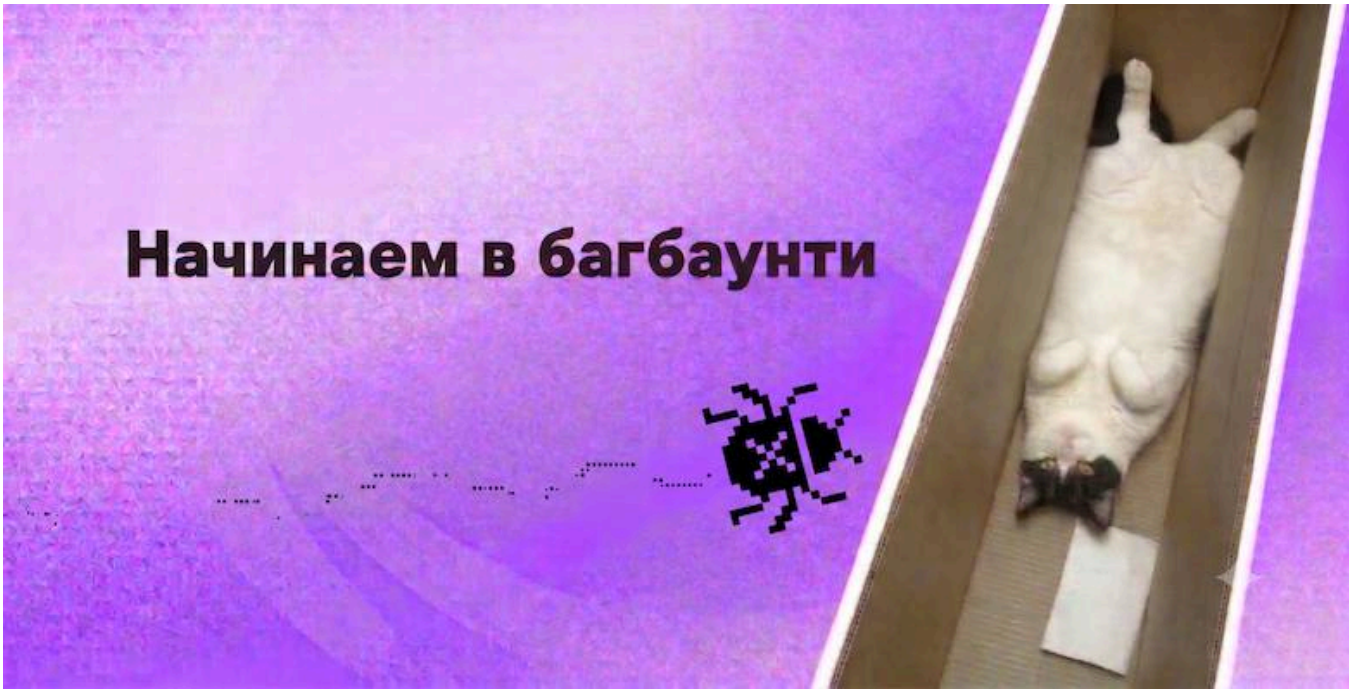
307.35

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Hello, everyone! My name is Sergey Bobrov, aka BlackFan. I work at Kaspersky Lab, where I analyze the security of mobile and web applications and take part in penetration tests and red team engagements. In my spare time, I also develop my own tools, conduct security research, and hunt for bugs on the Standoff Bug Bounty platform. Today, we will examine misconfigurations related to S3 object storage in practice and explore the problems that arise when integrating it into web applications through proxies such as nginx. The article is based on 10 laboratory environments, each illustrating different integration vulnerabilities: from classic stored XSS and bucket substitution to rewrite-rule bypasses and cache poisoning.

Everything discussed below can also be watched and heard in the video (also on VK Video).

> *This article is provided solely for informational purposes and is neither an instruction nor a call to commit unlawful acts. Its purpose is exclusively educational: to describe existing vulnerabilities that attackers could exploit, warn users, and provide recommendations for protecting personal information online. The author accepts no responsibility for the use of the published information. Remember to keep your data secure.*

S3: what a bug hunter needs to know

S3 (Simple Storage Service) is a service developed by Amazon to provide access to object storage in the AWS cloud. Because Amazon pioneered cloud services, its solutions became an industry standard. Today, “S3” refers not merely to a particular product, but also to a compatible API and an approach to storing data.

S3’s basic functionality can be described using just four HTTP requests:

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Creating or editing an object in a bucket (container) — `PUT`;

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Retrieving an object — `GET`;

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Deleting an object — `DELETE`;

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Listing a bucket's objects — `GET` to the bucket root.

In practice, developers often see S3 as "just a folder in the cloud," and this simplification creates considerable scope for misconfigurations and vulnerabilities when integrating S3 into web applications.

Authorization and request signatures

To make authorized requests, each user receives a pair of credentials: an Access Key and a Secret Key. A signature is generated as follows:

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All request components (method, Content-Type, date, and others) are converted into a canonical representation — a string separated by newline characters;

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An HMAC-SHA1 of this string is calculated;

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The result is Base64-encoded and passed in the `Authorization` header or query parameters.



There are two main signature formats: v2 (deprecated, but still used in many compatible solutions) and v4 (more complex). In my examples, I use v2 because it is simple and easy to illustrate. Signed URLs with an `Expires` parameter are used to provide temporary access to objects.

Two ways to integrate S3 into web applications

Through an S3 client in application code

Developers create a client, specify an endpoint and an Access Key/Secret Key pair, and then call the appropriate S3 methods. In this case, the attacker is limited to the elements passed to the client method: file contents, Content-Type, and filename.

Through a proxy (nginx)

S3 is often connected as storage for static assets through nginx using the `proxy_pass` directive. In this configuration, all S3 protocol functionality becomes available to an attacker because nginx does not filter methods and headers (apart from the path prefix). This is the most interesting case for us, because it is where most vulnerabilities arise.

Access control: ACL and Bucket Policy

Access control in S3 is implemented in two ways:

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S3 Bucket ACL (Access Control List) is a list of rules that defines permissions for a bucket or individual objects. The most dangerous mistake here is that developers often misinterpret the `AuthenticatedUsers` group. This does not mean users of a single account, but all authenticated S3 users worldwide (within the given cloud). This mistake has led to large-scale data leaks and sometimes to buckets being encrypted for ransom. For example, if you use Yandex Cloud, any other user can gain access to your bucket when this misconfiguration is present.

-

S3 Bucket Policy is a JSON document that lets you define bucket-level rules: permissions based on prefixes, conditions, and specific users. Unlike ACLs, the policy applies to the entire bucket, which simplifies administration but also opens up extensive opportunities for attacks when mistakes are made.

Identifying S3 Systems and Reconnaissance

Before attacking, it is important to understand exactly which S3 system we are dealing with. I compiled the S3 Fingerprint table, which collects characteristic headers and error pages for different implementations. It lets you identify S3 from the response to a request. The second repository is a dedicated extension for Burp Suite and Caido.

Recommendations:

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Send requests for an existing object and a nonexistent object, and compare the responses. This helps you understand how proxying works, identify which headers the frontend adds, and see whether internal S3 information is disclosed.

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Pay attention to the `X-Amz-Request-Id` header: its format (length, letter case, prefixes) often indicates a particular S3-compatible system (MinIO, Ceph RGW, Yandex Cloud, AWS).

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Use a Burp Suite extension (such as Hackvertor) to quickly generate signed requests with an invalid signature. This can reveal the canonical representation of the request, including the bucket name.

Practical Labs

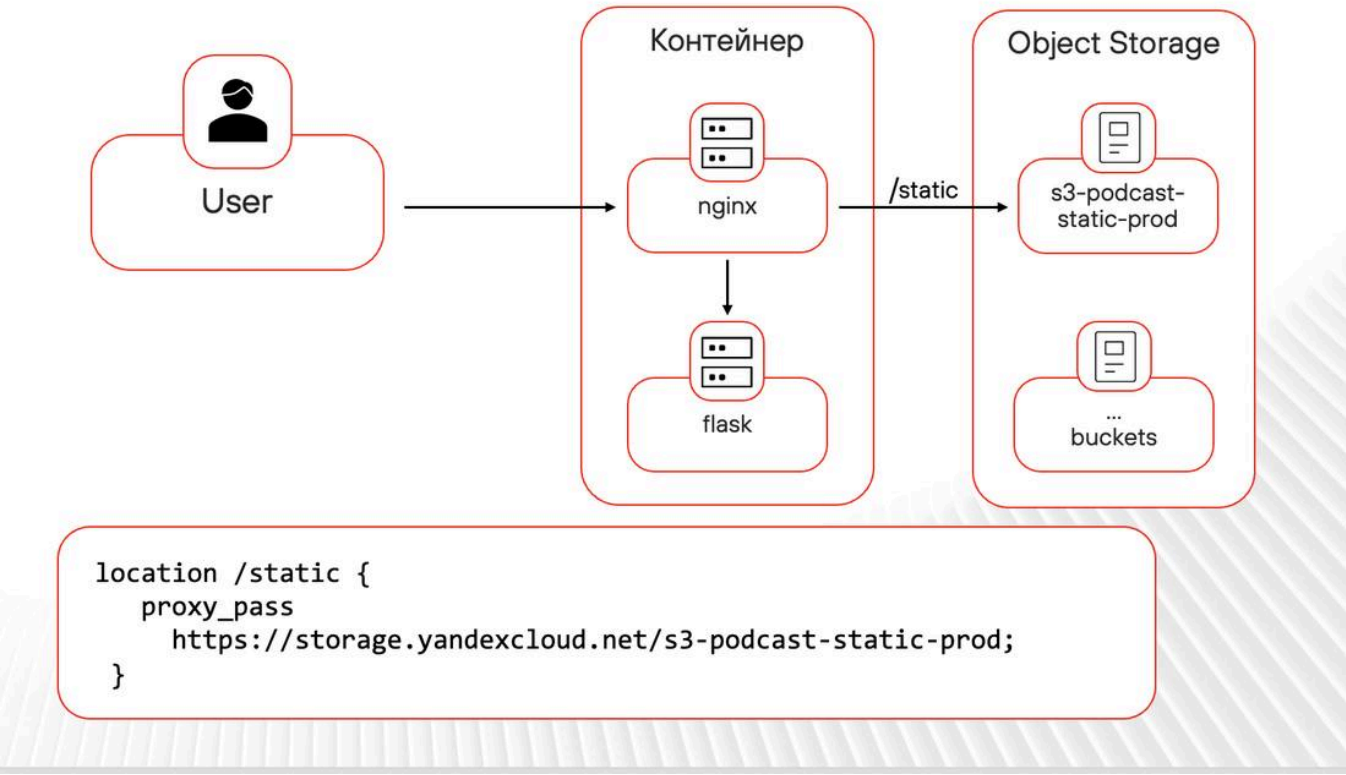
Let us move on to a series of labs. All the lab setups will look almost identical: an imageboard with pictures. However, despite the identical appearance, a difference of just one character in the site's configuration will lead to different errors, each exploited in its own way.

Errors in Proxying Rules

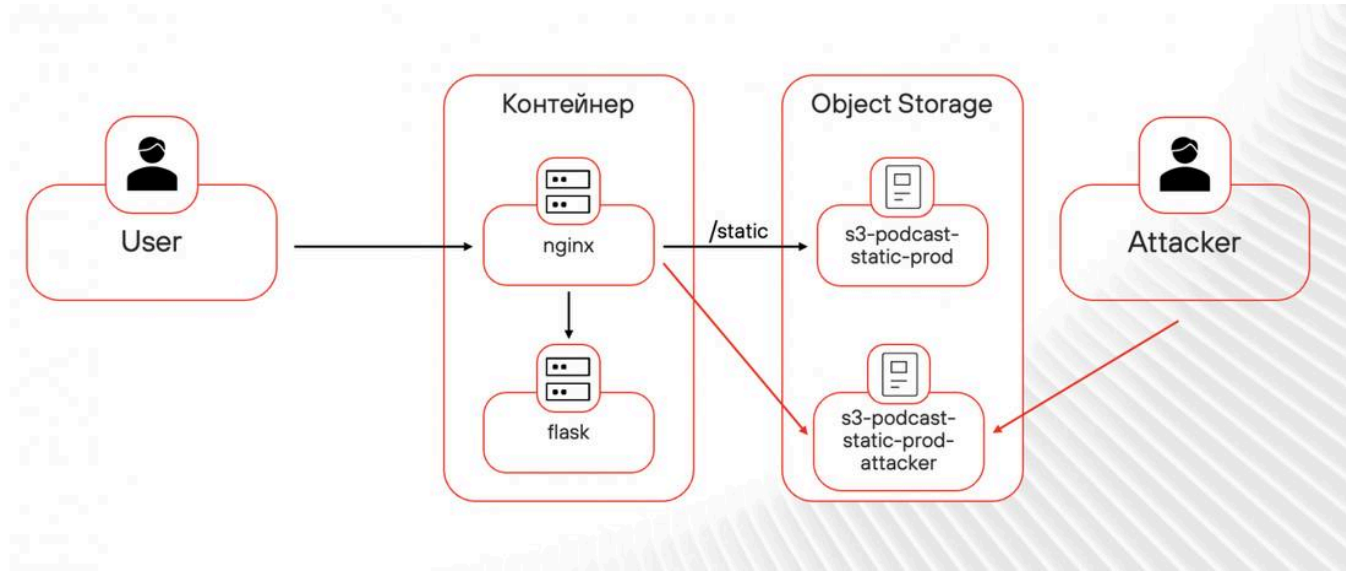
The proxy does not understand S3 semantics and works with the decoded and normalized path value. A one-character mistake in a rule creates a difference between what nginx sees and what is sent to storage, and the attacks below are built on precisely that discrepancy.

Bucket Substitution Through a Missing Trailing Slash

In this lab, the mistake is that the developer forgot to specify a trailing slash in the `/static` rule. If you request `/staticxxx`, the `static` prefix is replaced with the bucket name during proxying, and any additional data is concatenated with it. As a result, the request goes to a nonexistent bucket.



This means we can create a bucket with the original prefix, append some data to its name, upload any files, and access those files through the site we are interacting with. For example, after creating such a bucket in the attacker's cloud, we uploaded a file named xss.html containing a typical exploit. When users accessed it through the vulnerable site, the XSS was triggered.

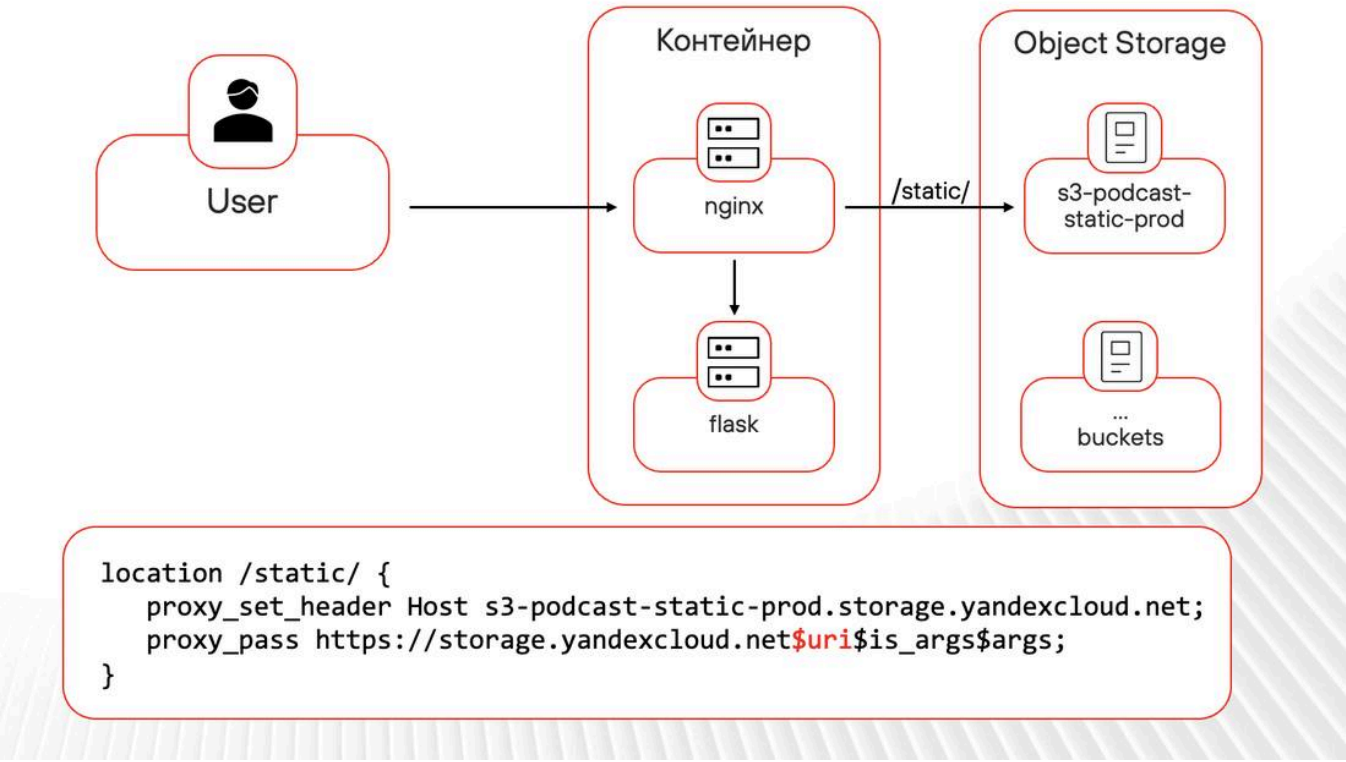


> 32:17–36:18 (solution demonstration) https://youtu.be/SwEpvGKsidc?t=1937

HTTP Request Splitting Through $uri and Cookie Theft

In this lab, the proxy rule uses the `$uri` variable, which contains the decoded request path. This allows an attacker to inject control characters into the path (for example, spaces and line breaks) and use **HTTP Request Splitting** to modify the request. The

vulnerability is detected as follows: adding a space and the letter H causes a `400` error (it is interpreted as the protocol version). Using this technique, the attacker replaces the `Host` header and redirects the request to an arbitrary bucket, loading an XSS exploit. Furthermore, by constructing a `PUT` request with the same vulnerability and sending it through XSS, part of the user's HTTP request can be written to the attacker's bucket together with their session cookies (including `HttpOnly`). A service account with the storage.uploader role is created to sign these requests.

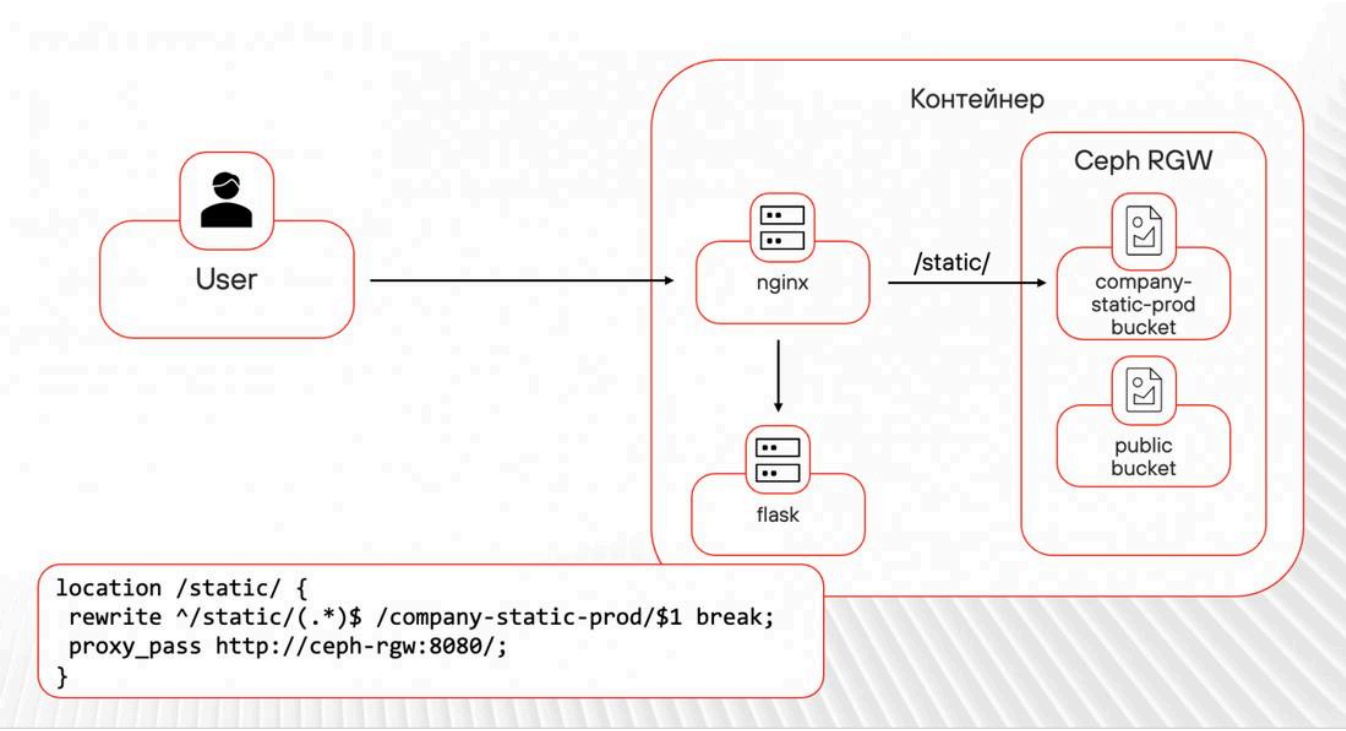


In brief, we found a way to carry out XSS attacks and also make the user's browser write sensitive data to a file in storage under our control.

> 36:19–50:11 (solution demonstration) https://youtu.be/SwEpvGKsidc?t=2178

Bypassing rewrite with %0A

This lab uses Ceph Rados Gateway, but here the bucket name is replaced using a regular expression—a rewrite rule. This configuration introduces the following problem: these rules often use the start-of-line (^) and end-of-line ($) characters in the regular expression. However, nginx works with the decoded and normalized path value, and if a line-break character (%0A) is placed after the static directory, the regular expression will not match. The correct bucket name will not be added to the HTTP request.



As a result, the static directory remains in the request, but `proxy_pass` with the `static` prefix is replaced with a single slash, and everything after static becomes the bucket name. Adding `%0A` in the middle of the object name produces a `NoSuchBucket` error. This means that the substitution rule did not work, and we can access an arbitrary bucket (for example, `test`) with the required suffix.

By enumerating bucket names in Intruder, we find the existing `public` bucket and construct a `PUT` request to create a file in the same way. As a result, we again gain the ability to execute XSS through proxying to S3.

> 55:47–59:40 (solution demonstration) https://youtu.be/SwEpvGKsidc?t=3347

Cache poisoning through query parameters

This lab has no main application—all traffic is proxied to S3 and cached by nginx for one minute. The cache key is formed only from the path, without taking `query` parameters into account. The mistake here is that the developers treat S3 as a static directory and do not account for parameters that change the response.

We send a request to `index.html` with the `?response-content-type=image/png` parameter. S3 returns `200 OK` with a modified Content-Type. nginx cached this response, and all subsequent users received `index.html` with the `image/png` type. This allows us to temporarily disrupt the site's functionality.

However, an attempt to use the `?acl` parameter, which is often available to unauthenticated users, failed: our bucket's permissions are configured correctly, so a 403 error is returned—such responses are not cached.

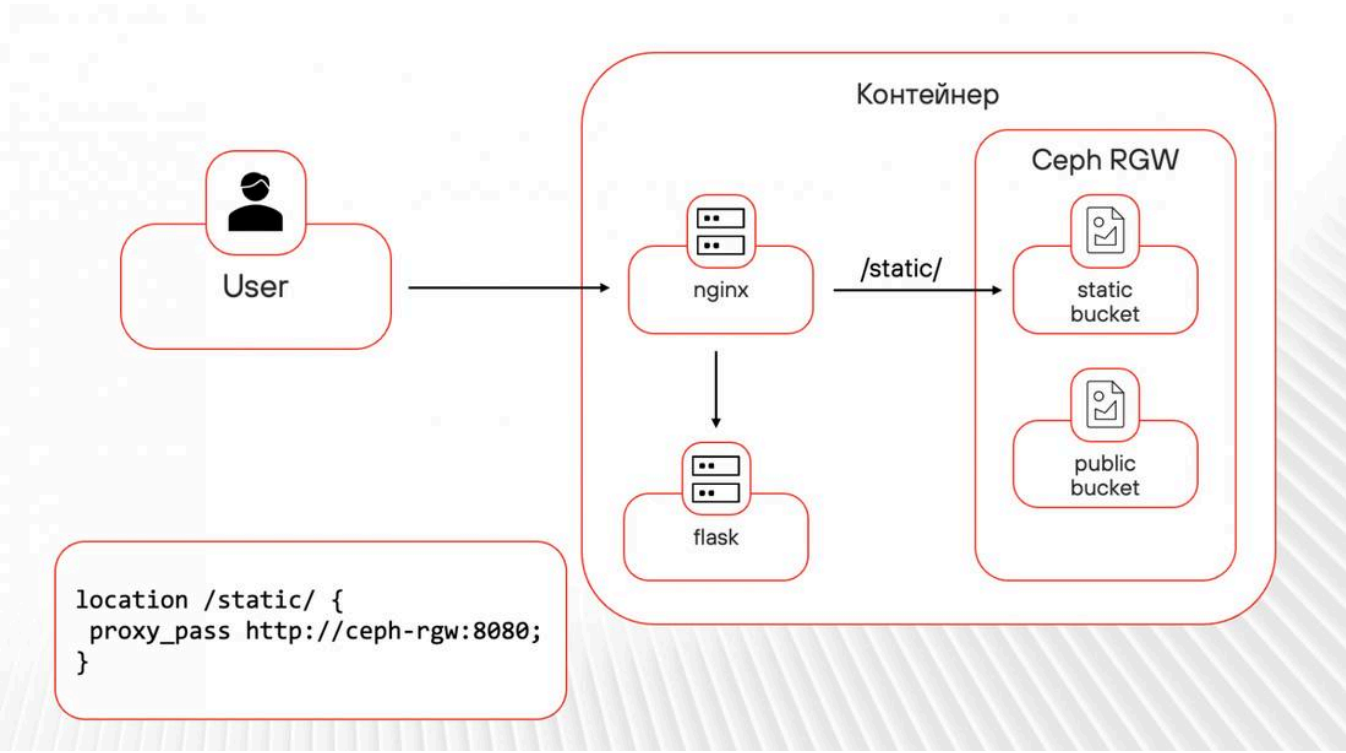
> 1:16:04–1:20:00 (solution demonstration) https://youtu.be/SwEpvGKsidc?t=4563

Characteristics of S3-compatible systems

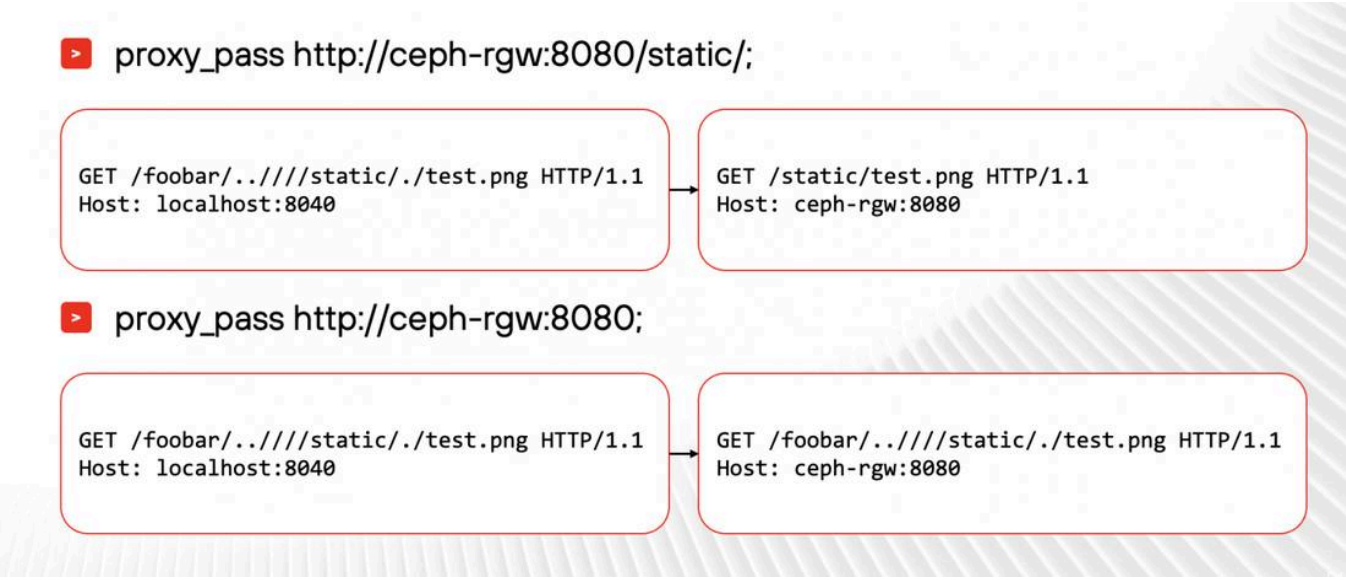
Here the proxy configuration looks quite reasonable, but S3-compatible systems behave differently from what the developer expects: they normalize paths differently and disclose internal information differently in error responses.

Path traversal through .. in Ceph RGW

This time we use Ceph Rados Gateway. It is a more advanced application than MinIO.



Here the proxy rules look quite reasonable, but the problem is that `proxy_pass` in nginx has two forms of proxying: with a path specified in the rule, or without one. To nginx, both requests look the same, but if no path is specified, the request is sent to the upstream server exactly as the user supplied it.



This means that, for S3 systems that do not normalize paths (unlike a filesystem), the `../` characters become a valid part of the object name. As a result, we can access arbitrary buckets, but with a fixed suffix.

Our goal is to access the internal `public` bucket, which is not intended for external interaction. When requesting an existing file, we see a new format for the `x-amz-request-id` header (it contains a long value with the `tx...` prefix), which corresponds to Ceph RGW.

After constructing an unnormalized path with `..`, we try to access another bucket and receive `NoSuchBucket` in response. We use Intruder to enumerate existing internal buckets. However, the `NoSuchBucket` errors have the same response status and size, so we need to add a rule to extract text from the content, using a regular expression or delimiters to extract the error code. In one of the results, we find a `NoSuchKey` error: this means that the `public` bucket exists.

We construct a `PUT` request, upload an arbitrary file to the misconfigured bucket, and access it through the vulnerable web application. Because the browser normalizes paths containing `..`, we had to encode one of the slashes so that the request would be sent correctly. In the end, we obtain XSS again.

> 50:10 – 55:41 (solution demonstration) https://youtu.be/SwEpvGKsidc?t=3010

Finding buckets through naming conventions

We established that the bucket name is passed in the `Host` header (virtual-host-style). A request with an invalid signature returns a detailed 403 error that discloses the canonical representation of the request, including the bucket name.

Because this is public object storage, developers try to use a consistent bucket naming convention. Knowing the company's convention, we can enumerate possible bucket names through Burp Intruder (attack type: Cluster bomb attack). We expect three types of responses:

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`404` — the bucket does not exist;

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`403` — it exists, but listing is forbidden;

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`200` — it exists and is misconfigured.

This is how we found an existing bucket belonging to another application that discloses sensitive data.

> 24:41–32:17 (solution demonstration) https://youtu.be/SwEpvGKsidc?t=1361

Abusing built-in protocol capabilities

The proxy is configured correctly, but the S3 protocol itself remains accessible from outside: listing, signed URLs, and Content-Type overrides are all built-in capabilities that become attack vectors when integrated carelessly.

Basic listing and file uploads (XSS)

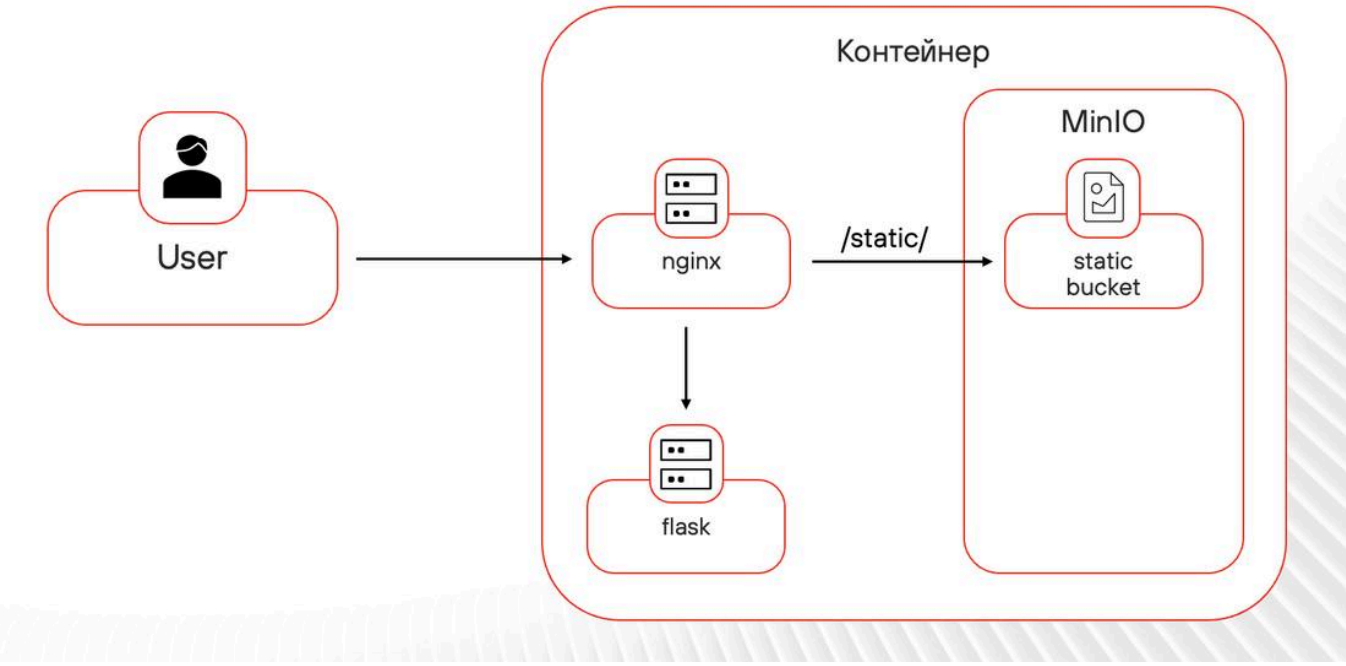
In this lab, the bucket was accessible only through a proxy with the `/static` prefix. We established that:

-

Bucket listing is enabled by default, and we can enumerate objects.

-

Uploading (PUT) is available to unauthenticated users.



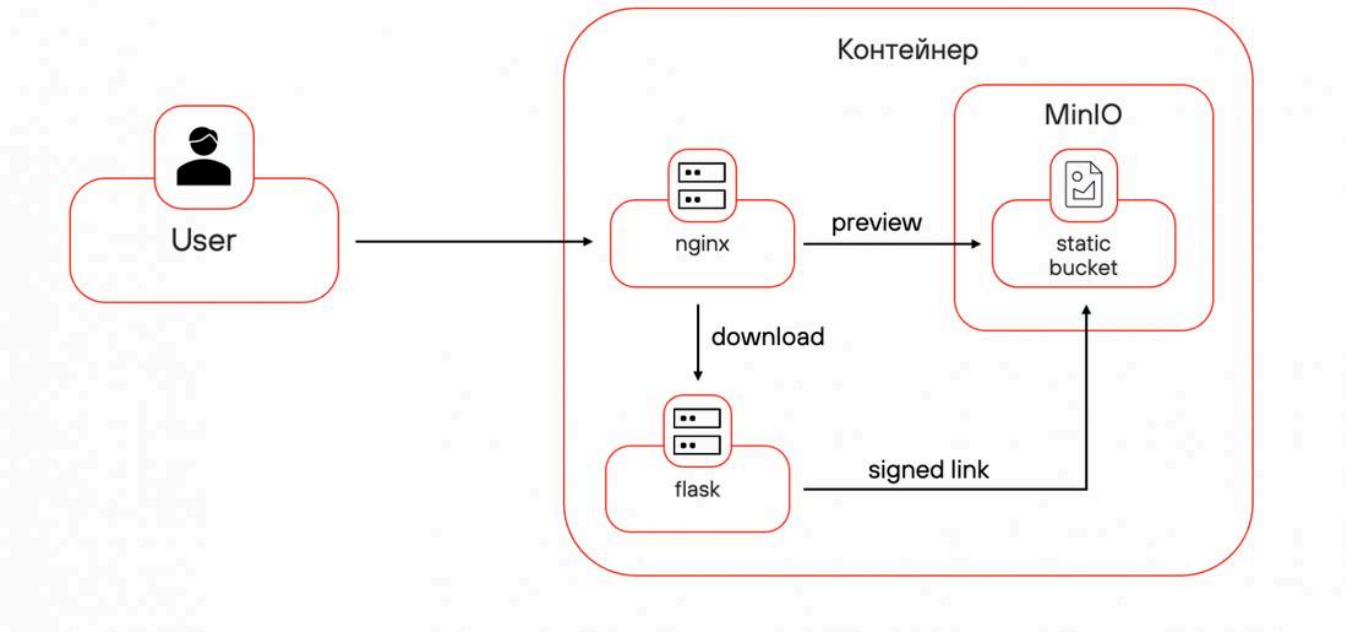
By uploading `test.html` with the `text/html` Content-Type, we gained the ability to execute arbitrary HTML/JS in the site's context. This is classic stored XSS.

If we see this misconfiguration and, in addition to static files, the storage contains JavaScript loaded by the homepage, we can overwrite it and inject a malicious script for all visitors. Doing this on production systems is not recommended, because it can easily break the site's functionality.

> 12:04-21:10 (solution demonstration) https://youtu.be/SwEpvGKsidc?t=725

Signed URLs and bucket listing

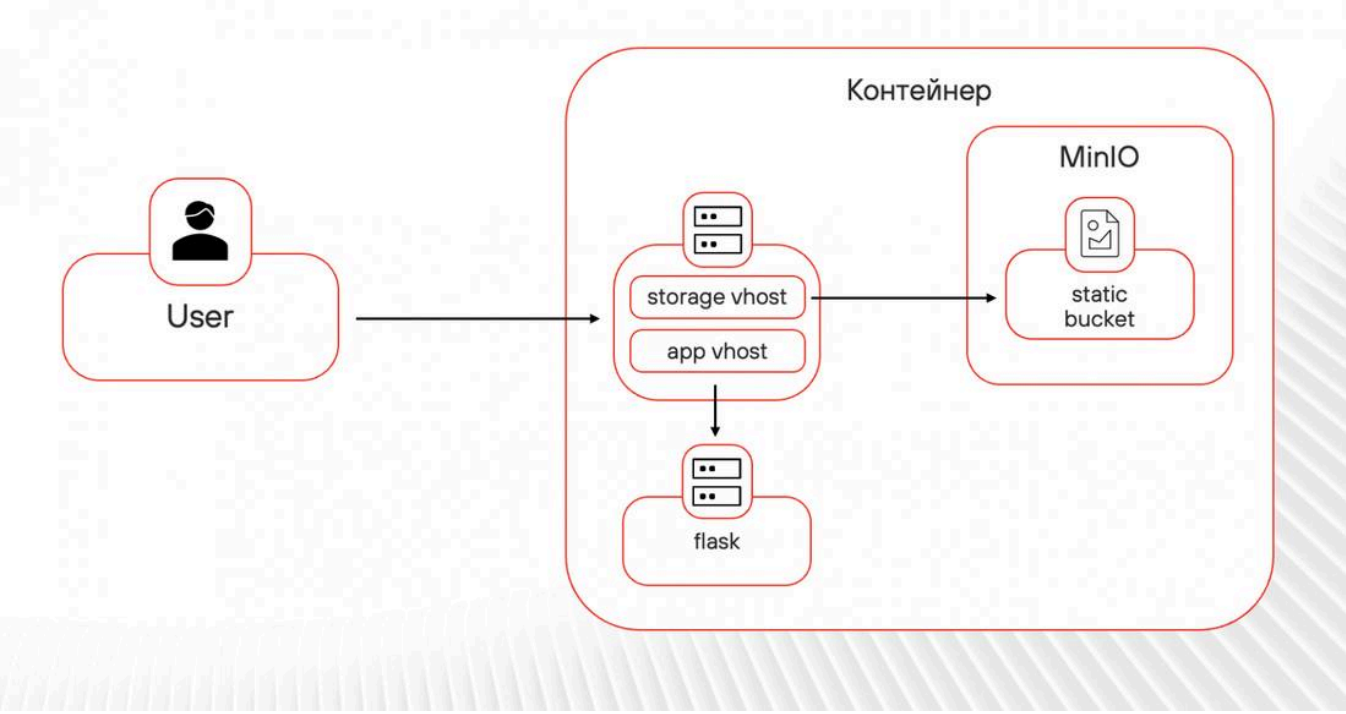
This lab implemented downloads through signed URLs. Here, the path to the file being downloaded is passed to the `/download` endpoint. If, instead of an object path, we specify a slash (/), for example, the server returns a signed URL pointing to the bucket root. In an S3 system, such a URL provides a list of all objects in the bucket, for which we can then generate signed URLs and access their contents.



> 21:12 - 24:03 (solution demonstration) https://youtu.be/SwEpvGKsidc?t=1272

Service Worker and content substitution

In this scenario, the application used two separate hosts: one interacted with the S3 bucket, while the other served the main application (flask). This separation is often used to isolate user content from the main application. Our goal is to make a legitimate user receive the attacker's file when attempting to download their own file.



After creating an attacker account in a separate browser to ensure the exploit works correctly, we upload an XSS payload, for example. This application requires the `.webp` suffix. If we try to upload an HTML file with the `.webp` extension, downloading it gives us

an invalid image because the application automatically supplies the corresponding `Content-Type`.

To change the Content-Type to `text/html` and make our uploaded HTML file executable, we use the `response-content-type` parameter in the GET request. However, because the subdomain is isolated, this does not provide access to the main application's cookies. A Service Worker helps us here: we upload a JavaScript file with the correct Content-Type to the subdomain and register it as a Worker using a separate HTML file. It now intercepts all of the user's fetch requests on this subdomain, sends information (file paths) to our collaborator, and replaces responses with a link to our malicious file (for example, a malicious image).

> 1:06:20–1:16:03 (solution demonstration) https://youtu.be/SwEpvGKsidc?t=3980

When vulnerability classes are chained together

A real attack rarely comes down to a single mistake. The final lab shows how an application-side leak and a standard protocol feature combine into a single attack vector.

Key disclosure through Django debug mode

In this lab, the application is based on Django and runs in debug mode. Our goal here is to access the `secret` bucket and its `secret.txt` file, which should not be accessible externally.

When we deliberately triggered an error, a complete traceback was displayed, including environment variable values. Automatic masking is present, but our S3 credentials are stored under the short names `S3_SK` and `S3_AK`, which allowed us to disclose the credentials used to interact with S3 on this site. With this key, we can sign any S3 requests.

Although the proxy was configured correctly, we used the special `X-Amz-Copy-Source` header to copy an object from the internal secret bucket into a publicly accessible one. This allowed us to bypass the proxy's restrictions and extract data that we could not access directly.

> 59:41–1:06:14 https://youtu.be/SwEpvGKsidc?t=3580 (solution demonstration)

Conclusion

Researching S3 integration vulnerabilities is important because even one missing character in an nginx configuration can compromise users' data and sessions. Our material shows that most problems arise not in S3 itself, but during its integration—a blind spot that developers often overlook. For a bug hunter, knowing these scenarios provides a ready-made set of vectors for real attacks, from simple bucket listing to stealing HttpOnly cookies while bypassing restrictions.

Tags:

- s3

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